THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 12: **Surface Areas and Volumes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A solid cylinder of radius 7 cm and height 8 cm is melted and recast into a cone of the same base radius. The height of the cone is: 1

(a) 16 cm **(b)** 48 cm **(c)** 8 cm **(d)** 24 cm

2. A solid cone and a solid hemisphere have equal base radii 10 cm and equal volumes. The height of the cone is: 1

(a) 30 cm **(b)** 20 cm **(c)** 15 cm **(d)** 10 cm

3. The radii of two solid spheres are in the ratio 3 : 5. The ratio of their surface areas is: 1

(a) 27 : 125 **(b)** 9 : 25 **(c)** 3 : 5 **(d)** 18 : 25

4. The radii of two solid spheres are in the ratio 3 : 4. The ratio of their volumes is: 1

(a) 9 : 16 **(b)** 27 : 32 **(c)** 3 : 4 **(d)** 27 : 64

5. A solid cylinder of base radius 7 cm and height 12 cm is melted and recast into identical solid cones, each of base radius 7 cm and height 6 cm. The number of cones obtained is: 1

(a) 2 **(b)** 12 **(c)** 6 **(d)** 10

6. A hemispherical bowl has outer radius 15 cm and inner radius 9 cm. The volume of material used to make the bowl is: 1

(a) 16632 cm^3 **(b)** 2772 cm^3 **(c)** 11088 cm^3 **(d)** 5544 cm^3

7. The largest possible solid sphere is carved out of a cube of edge 63 cm. The volume of the material wasted is: 1

(a) 250047 cm^3 **(b)** 130977 cm^3 **(c)** 184558.5 cm^3 **(d)** 119070 cm^3

8. A solid sphere of radius 31.5 cm fits exactly inside a cylinder (it touches the curved surface and both bases). The volume of the space in the cylinder not occupied by the sphere is: 1

(a) 130977 cm^3 **(b)** 327442.5 cm^3 **(c)** 65488.5 cm^3 **(d)** 196465.5 cm^3

9. **Assertion (A):** When a solid metallic sphere is melted and recast into a cylinder, its surface area remains the same. 1

Reason (R): When a solid is melted and recast, its volume remains the same.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The slant height of a cone of base radius 6 cm and height 8 cm is 10 cm. 1
Reason (R): The volume of a sphere of radius r is $\frac{4}{3}\pi r^3$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A hemispherical bowl has outer radius 14 cm and inner radius 3.5 cm. Find the volume of material used to make the bowl. 2

12. A solid metallic sphere of radius 9 cm is melted and drawn into a wire of circular cross-section of radius 1 cm. Find the length of the wire. 2

13. A solid sphere of radius 12 cm is completely dropped into water in a cylindrical vessel of base radius 8 cm, and no water spills. Find the rise in the water level. 2

14. A hemispherical bowl has outer radius 13.5 cm and inner radius 6 cm. Find the volume of material used to make the bowl. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A solid metallic cylinder of base radius 12 cm and height 24 cm is melted and recast into solid spheres of radius 6 cm. Find the number of spheres obtained. 3

16. A solid metallic cone of base radius 5 cm and height 20 cm is melted and recast into a solid sphere. Find the radius of the sphere. 3

17. A solid cylinder of base radius 8 cm and height 9 cm is melted and recast into identical solid cones, each of base radius 4 cm and height 6 cm. Find the number of cones obtained. 3

OR

A solid metallic cylinder of base radius 12 cm and height 16 cm is melted and recast into solid spheres of radius 6 cm. Find the number of spheres obtained. 3

18. A solid metallic cone of base radius 10 cm and height 40 cm is melted and recast into a solid sphere. Find the radius of the sphere. 3

OR

A solid cylinder of base radius 5 cm and height 24 cm is melted and recast into identical solid cones, each of base radius 5 cm and height 8 cm. Find the number of cones obtained. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A circus tent is made of a cylindrical lower part topped by a conical roof. The base radius of both parts is 7 m, the cylindrical part is 6 m high and the conical part has a vertical height of 24 m. (Take $\pi = \frac{22}{7}$.) 5
(i) Find the slant height of the cone.
(ii) Find the curved surface area of the cone.
(iii) Find the curved surface area of the cylindrical part.
(iv) Find the total canvas needed and its cost at ₹ 50 per square metre.

20. A capsule is a cylinder of radius 21 cm and length 45 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 5

OR

A hemispherical depression of diameter 147 cm, the largest possible, is scooped out of the top face of a cubical block of edge 147 cm. Find the volume of the remaining block. 5

END OF QUESTION PAPER Total: 40 Marks